Some important distributions

A. Discrete distributions

1. Bernoulli distribution
2. Binomial distribution
3. Poisson distribution
4. Hypergeometric distribution
5. Geometric distribution
6. Negative binomial

B. Continuous distributions

1. Uniform distribution
2. Exponential distribution
3. Gamma distribution
4. Normal distribution

Bernoulli distribution

Bernoulli trial

A trial that has two possible outcomes is known as a Bernoulli trial. The two outcomes are called ‘success’ (probability p) and ‘failure’ (probability $1-p$).

Examples

1. A fair coin is tossed. Here, head (or tail) is a success. $P(H) = p = 0.5$.

2. A fair dice is tossed. $P(6) = p = 1/6$.

3. A patient is tested for COVID-19. $P(+) = p = 0.05$ (say).

Bernoulli random variable

Let a Bernoulli trial be performed <u>once</u>. Then, $S = \{s,\ f\}$. Let X be the number of successes obtained. Then, X is a Bernoulli random variable with possible values ‘0’ and ‘1’. It has the following pmf:

x	0	1
$P(X = x)$	$1 - p$	p

We can write the above pmf as follows:

$P(X = x) = p^x(1 - p)^{1-x}\,; x = 0, 1.$

Note

Here, p is the 'parameter' of the probability distribution. It characterizes the probability distribution (population). We write,

$X \sim \text{Bernoulli}(p)$.

That is, X follows Bernoulli distribution with parameter p.

Mean and variance

$E(X) = 0 \times (1 - p) + 1 \times p = p$

$E(X^2) = 0^2 \times (1 - p) + 1^2 \times p = p$

$V(X) = p - p^2 = p(1 - p)$

Note that, mean $>$ variance.

Also, $V(X) \leq 0.25$. (Why?)

Example

In a multiple-choice question (MCQ), there are four choices. A student does not know the correct answer and selects one of the choices at random. Let X be the number of correct answers (out of 1 question). Then

$X \sim \text{Bernoulli}(p = 0.25)$.

Binomial distribution

Binomial experiment

A binomial experiment consists of n <u>independent</u> Bernoulli trials, where n is a positive integer. The probability of success (p) is <u>same</u> in each trial.

Example

1. A fair coin is tossed 5 times. Here, $n = 5$ and $p = 0.5$.

2. A coin (probability of head 0.7) is tossed 10 times. Here, $n = 10$ and $p = 0.7$.

Binomial random variable

Let n <u>independent</u> Bernoulli trials be performed, each having probability of success p. Let X be the number of successes obtained. Then, X is a binomial random variable with possible values $0, 1, 2, \cdots, n$. It has the following pmf:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}\ ; x = 0, 1, 2, \cdots, n.$$

Note

The distribution has two parameters n and p. We write

$X \sim \text{binomial}(n,\ p)$.

That is, X follows binomial distribution with parameters n and p.

Explanation of the pmf

Let 5 independent Bernoulli trials be performed, each having probability of success p. The sample space is as follows:

$$S = \{sssss, ssssf, sssfs, sssff, \cdots, sffss, \cdots, ffsss, \cdots, fffff\}$$

Since the trials are independent, we have

$$P(sssss) = p \times p \times p \times p \times p = p^5$$
$$\vdots$$
$$P(sssff) = p \times p \times p \times (1-p) \times (1-p) = p^3(1-p)^2$$
$$\vdots$$
$$P(sffss) = p \times (1-p) \times (1-p) \times p \times p = p^3(1-p)^2$$
$$\vdots$$
$$P(fffff) = (1-p) \times \cdots \times (1-p) = (1-p)^5$$

Probability that we will get 3 successes:

$$\begin{aligned} P(X = 3) &= P(sssff \text{ or } ssffs \text{ or } \cdots \text{or } ffsss) \\ &= p^3(1-p)^2 + p^3(1-p)^2 + \cdots + p^3(1-p)^2 \\ &= \binom{5}{3} p^3(1-p)^2 \end{aligned}$$

Note

$(p + (1 - p))^n = 1$

If we expand the left-hand-side of the above equation (binomial expansion), we get the sum of all the probabilities of binomial distribution, which equals one.

Note

Let $X \sim$ binomial$(n,\ p)$. Then

$X = Y_1 + Y_2 + \cdots + Y_n$

where $Y_i =$ number of success in the ith trial, $i = 1, 2, \cdots, n$. That is, $Y_i \sim \text{Bernoulli}(p)$. Thus, a $\text{binomial}(n,\ p)$ random variable is the sum of n independent $\text{Bernoulli}(p)$ random variables.

Mean and variance

$E(X) = np$

$V(X) = np(1 - p)$

Variance is smaller than the mean.

Exercise

In an MCQ test, there are 10 questions each with 4 choices. Smith chooses all the answers at random. What is the probability that the number of correct answers will be (i) exactly 2 (ii) less than 2 (iii) 2 or less?

Solution

Let $X =$ number of correct answers. Then

$X \sim$ binomial$(n = 10,\ p = 0.25)$.

(i) $P(X = 2) = P(2) = \binom{10}{2} 0.25^2 (1 - 0.25)^{10-2} = 0.2816$

(ii) $P(X < 2) = P(0) + P(1) = 0.2440$

(iii) $P(X \leq 2) = P(0) + P(1) + P(2) = 0.5256$

Exercise

In an MCQ test, there are 10 questions each with 4 choices. Neither Smith nor Jones studied for the test. They independently chose all the answers at random. What is the probability that their total number of correct answers will be 5?

Solution

Let X_1 = number of correct answers by Smith.

and X_2 = number of correct answers by Jones.

Here, X_1 and X_2 are independent.

Let $Y = X_1 + X_2$. Then

$Y \sim$ binomial$(n = 20,\ p = 0.25)$.

$P(Y = 5) = P(5) = \binom{20}{5} 0.25^5 (1 - 0.25)^{20-5} = 0.2023$

Note

The above problem is an example of 'additive property' of binomial distribution. If $X_1 \sim$binomial$(n_1,\ p)$ and $X_2 \sim$binomial$(n_2,\ p)$, and if X_1 and X_2 are independent, then

$Y = X_1 + X_2 \sim$ binomial$(n_1 + n_2,\ p)$.

Poisson distribution

A Poisson distribution is appropriate for 'count data' which are obtained by counting the number of occurrences in an interval of time or in an area.

Examples

1. Number of accidents at a particular spot in a week
   $(X = 0, 1, 2, \cdots)$
2. Number of pens required in a month
   $(X = 0, 1, 2, \cdots)$
3. Number of jackfruit trees in every square kilometer
   $(X = 0, 1, 2, \cdots)$

Poisson pmf

A Poisson random variable X has the following pmf:

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \ ; \ \ x = 0, 1, 2, \cdots ; \ \ \lambda > 0.$$

Here, λ is the parameter of the distribution. We write

$X \sim \text{Poisson}(\lambda)$.

Mean and variance

$E(X) = \lambda$

$V(X) = \lambda$

Note that mean = variance.

Exercise

The average number of accidents at a particular spot in a week is 3.5. What is the probability that the number of accidents in the next week will be (i) zero (ii) exactly one (iii) more than one?

Solution

Let $X =$ number of accidents in the next week. Then

$X \sim \text{Poisson}(\lambda = 3.5)$.

(i) $P(X = 0) = P(0) = \dfrac{e^{-3.5}3.5^{0}}{0!} = 0.0302$

(ii) $P(X = 1) = P(1) = 0.1057$

(iii) $P(X > 1) = 1 - P(0) - P(1) = 0.8641$

Exercise

On an average, 3.5 customers enter a shop between 10:00-11:00 and 2.5 customers enter the shop between 11:00-12:00. What is the probability that exactly 5 customers will enter the shop between 10:00-12:00 tomorrow?

Solution

Let X_1 = number of customers between 10:00-11:00.

and X_2 = number of customers between 11:00-12:00.

Here, X_1 and X_2 are independent.

Let $Y = X_1 + X_2$. Then

$Y \sim \text{Poisson}(\lambda = 3.5 + 2.5 = 6)$.

$$P(Y = 5) = P(5) = \frac{e^{-6} 6^5}{5!} = 0.1606$$

Note

The above problem is an example of 'additive property' of Poisson distribution. If $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$, and if X_1 and X_2 are independent, then

$Y = X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Binomial-Poisson relationship

Theorem: Let $X \sim \text{binomial}(n,\ p)$. If $n \to \infty$ and $p \to 0$ so that np is finite, then

$X \sim \text{Poisson}(\lambda = np)$, approximately.

Explanation

Let X be a binomial variable. If n is very large, p is very small and np is neither large nor small, then X follows Poisson distribution approximately. In that case, we can use either binomial or Poisson distribution to solve the problem.

Exercise

Batteries of a particular brand malfunctions with probability 0.001. Calculate the probability that at least one out of 1000 batteries will malfunction.

Solution

Method 1

Let X = number of batteries that will malfunction. Then

$X \sim \text{binomial}(n = 1000,\ p = 0.001)$.

$P(X \geq 1) = 1 - P(0) = 0.6323$

Method 2

$X \sim \text{binomial}(n = 1000,\ p = 0.001)$.

Here, n is large and p is small. Also, $np = 1$ (neither large nor small).

$X \sim \text{Poisson}(\lambda = np = 1)$, approximately.

$P(X \geq 1) = 1 - P(0) = 0.6321$

Hypergeometric distribution

This distribution is appropriate when we are sampling <u>without replacement</u> from a finite population.

Example

A box contains 20 mobile phones of which 12 are good and 8 are defective. You took 5 phones at random from the box <u>without</u> replacement. Let X denote the number of phones that are good out of the 5 taken. ($X = 0, 1, 2, 3, 4, 5$).

Example

There are 30 fish in a pond of which 20 are tilapia and 10 are catfish. A total of 7 fish are caught with a fishing rod. Let X denote the number of tilapias caught.

Hypergeometric pmf

Let the population have N items of which M are 'good' and $N - M$ are 'bad'. Let n items be drawn at random without replacement. Let X be the number of 'good' items drawn. Then, X has the following pmf:

$$P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}\ ;\ \ x = 0, 1, 2, \cdots, n$$

[when $n < M$ and $n < N - M$.]

Here, $X \sim \text{hypergeometric}(N, M, n)$.

Exercise

There are 20 tilapia and 10 catfish in a small pond. You caught 5 fish from the pond with a net. What is the probability that exactly 2 of the fish are tilapia?

Solution

$$P(X=2)=\frac{\binom{20}{2}\binom{10}{3}}{\binom{30}{5}}=0.1600$$

Comparison with binomial distribution

Let a box contain 10 phones: 6 good and 4 bad. Let us draw 3 phones at random without replacement.

Each trial (draw) has two possible outcomes: good and bad (Bernoulli trial). However, since draws are done without replacement, probability of good (success) changes from draw to draw.

For the 1st draw: $P(G)=6/10$.

For the 2nd draw: $P(G)=6/9$ or $5/9$
(depending on the outcome of 1st draw)

and so on. Thus, conditions of binomial distribution are NOT fulfilled. If the draws are done with replacement, we will have a binomial setup.

Hypergeometric to binomial

If the population is large, there is almost no difference between sampling with replacement and sampling without replacement.

Let a box contain 10000 phones: 6000 good and 4000 bad. Let us draw 3 phones at random without replacement.

For the 1st draw: $P(G)=6000/10000$.

For the 2nd draw: $P(G)=6000/9999$ or $5999/9999$

All three numbers are almost equal. Thus,

When $N\to\infty$, hypergeometric(N,M,n) reduces to binomial$\left(n,p=\frac{M}{N}\right)$.

Exercise

There are 2000 tilapia and 1000 catfish in a large pond. You caught 5 fish from the pond with a net. What is the probability that exactly 2 of the fish are tilapia?

Solution

Let X be the number of tilapias caught.

Method 1

$X \sim \text{hypergeometric}(N = 3000, M = 2000, n = 5)$.

$$P(X = 2) = \frac{\binom{2000}{2}\binom{1000}{3}}{\binom{3000}{5}} = 0.16458$$

Method 2

$$X \sim \text{binomial}\left(n = 5,\ \ p = \frac{2000}{3000} = \frac{2}{3}\right)$$

$$P(X = 2) = \binom{5}{2}(2/3)^2(1 - 2/3)^{5-2} = 0.16461$$

Geometric distribution

This distribution is appropriate when independent Bernoulli trials are performed until we get the first success. Each trial has the same probability of success. We are interested in the number of failures obtained before the success. Alternatively, we are interested in the number of trials required to get the first success.

Example

A fair coin is tossed repeatedly until we get a tail. Let X denote the number of heads observed before getting the first tail. Here, $X = 0, 1, 2, \cdots$. Let Y denote the total number of trials required. Then, $Y = 1, 2, 3, \cdots$. Note that $Y = X + 1$.

Example

Fishes are caught one after another with a fishing rod from a large pond until we get a catfish. Let X denote the number of other fishes caught before getting the catfish. Here, $X = 0, 1, 2, \cdots$. Let Y denote the total number fishes (including the catfish). Then, $Y = 1, 2, 3, \cdots$.

Geometric pmf

Let independent Bernoulli trials be performed until we get the first success. Then, $S = \{s, fs, ffs, fffs, \cdots\}$. Let The probability of success in each trial be p. Let X denote the number of failures observed before getting the success. Then,

$P(X = x) = (1 - p)^x \, p \; ; \;\; x = 0, 1, 2, \cdots.$

We write, $X \sim \text{geometric}(p)$

Mean and variance

$$E(X) = \frac{1 - p}{p}$$

$$V(X) = \frac{1 - p}{p^2}$$

Exercise

A fair die is thrown until '1' occurs. What is the probability that exactly 6 <u>throws</u> will be required?

Solution

Let X be the number of <u>failures</u> before getting '1'.

$X \sim \text{geometric}(p = 1/6)$.

$$P(X = 5) = \left(1 - \frac{1}{6}\right)^5 \frac{1}{6} = 0.0670$$

Note

$$\begin{aligned} P(X \geq k) &= P(X = k) + P(X = k + 1) + \cdots \\ &= (1 - p)^k \, p + (1 - p)^{k+1} \, p + \cdots \\ &= (1 - p)^k \end{aligned}$$

Memoryless property

$P(X \geq k + m \mid X \geq k) = P(X \geq m)$

Proof:

$P(X \geq k + m \mid X \geq k)$

$= \dfrac{P(X \geq k + m)}{P(X \geq k)}$

$= \dfrac{(1-p)^{k+m}}{(1-p)^k}$

$= (1-p)^m$

$= P(X \geq m)$

Explanation

Suppose, you are catching fishes one after another with a fishing rod until you get a catfish. Let X denote the number of other fishes caught (number of failures) before getting the catfish. According to the memoryless property:

$P(X \geq 10 + 5 \mid X \geq 10) = P(X \geq 5)$

Suppose, you already have 10 failures, but no catfish is caught yet. Then, the probability that you will have at least 5 more failures is same as the probability of at least 5 failures when you just started. That is, the system has 'forgotten' that you already have10 failures.

Negative binomial distribution

This distribution is appropriate when independent Bernoulli trials are performed until we get r successes. Each trial has the same probability (p) of success. We are interested in the number of failures obtained before getting r successes.

Examples

A fair coin is tossed repeatedly until we get 3 tails. Let X denote the number of heads observed before getting 3 tails. Here, $X = 0, 1, 2, \cdots$. Let Y denote the total number of trials required. Then, $Y = 3, 4, 5, \cdots$. Note that $Y = X + 3$.

Fishes are caught one after another with a fishing rod from a large pond until we get 5 catfish. Let X denote the number of other fishes caught before getting 5 catfish. Here, $X = 0, 1, 2, \cdots$.

Negative binomial pmf

Let independent Bernoulli trials be performed until we get r successes. Let the probability of success in each trial be p. Let X denote the number of failures observed before getting r successes. Then,

$$P(X = x) = \binom{x + r - 1}{r - 1} p^r (1 - p)^x \; ; \;\; x = 0, 1, 2, \cdots.$$

We write, $X \sim$ negative binomial(r, p).

Explanation

If we have x failures before getting r successes, the total number of trials is $x + r$. The last trial, i.e., the $(x + r)$th trial, has yielded the rth success, while the previous $x + r - 1$ trials have yielded $r - 1$ successes. Thus,

$P(X = x)$

$= P(r - 1 \text{ successes in } x + r - 1 \text{ trials and a success in the last trial})$

$= P(r - 1 \text{ successes in } x + r - 1 \text{ trials}) \times P(\text{a success in the last trial})$

$$= \binom{x + r - 1}{r - 1} p^{r-1} (1 - p)^{(x+r-1)-(r-1)} \times p$$

$$= \binom{x + r - 1}{r - 1} p^r (1 - p)^x$$

Note

Let $X \sim$ negative binomial$(r,\ p)$. Then

$$X = Y_1 + Y_2 + \cdots + Y_r$$

where $Y_1 =$ number of failures before the 1st success, $Y_2 =$ number of failures between the 1st success and the 2nd success, and so on. That is, $Y_i \sim$ geometric(p), $i = 1, 2, \cdots, r$. Thus, a negative binomial$(r,\ p)$ random variable is the sum of r independent geometric(p) random variables.

Mean and variance

$$E(X) = \frac{r(1 - p)}{p}$$

$$V(X) = \frac{r(1 - p)}{p^2}$$

Exercise

A fair dice is thrown until ‘1’ occurs 3 times. What is the probability that exactly 7 throws will be required?

Solution

Let X be the number of failures before ‘1’ occurs 3 times.

$X \sim$ negative binomial$(r = 3,\ p = 1/6)$.

$$P(X = 4) = \binom{4 + 3 - 1}{3 - 1} \left(\frac{1}{6}\right)^3 \left(1 - \frac{1}{6}\right)^4 = 0.0335$$